

Session 3

0031. Declare variables for a Flipkart product: productName (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.

```
#include <stdio.h>
```

```
int main() {
```

```
    char productName[] = "Samsung Galaxy S24";
```

```
    float price = 74999.50f;
```

```
    double rating = 4.5;
```

```
    printf("Product Name: %s (String)\n", productName);
```

```
    printf("Price: %.2f (Float)\n", price);
```

```
    printf("Rating: %.1f (Double)\n", rating);
```

```
    return 0;
```

```
}
```

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2. Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.

Constraint: The GST rate must not be changeable after its initial assignment.

```
#include <stdio.h>
```

```
int main() {
```

```
    const float GST_RATE = 18.0f;
```

```
    float basePrice = 500.0f;
```

```
    float gstAmount;
```

```
    float finalPrice;
```

```
    gstAmount = basePrice * GST_RATE / 100;
```

```
    finalPrice = basePrice + gstAmount;
```

```
    printf("Base Price: %.2f\n", basePrice);
```

```
    printf("GST Rate: %.2f%%\n", GST_RATE);
```

```
    printf("GST Amount: %.2f\n", gstAmount);
```

```
    printf("Final Price: %.2f\n", finalPrice);
```

```
    return 0;
```

```
}
```

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3. Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.

```
#include <stdio.h>
```

```
int main() {
```

```
    char playlistName[] = "My Favorite Songs";
```

```
    int totalSongs = 25;
```

```
    float averageDuration = 3.5f;
```

```
    printf("My Spotify playlist is \"%s\" with %d songs, and the average song  
duration is %.1f minutes.\n",
```

```
        playlistName, totalSongs, averageDuration);
```

```
    return 0;
```

```
}
```

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4. Identify and correct the invalid variable names from this list: 1stPlayer, player_age, \$score, total-marks, userName.

Hint: Apply standard variable naming rules for your programming language.

Variable name	Reason	Correct name
1stPlayer	Invalid – cannot start with a number	firstPlayer
player_age	Valid	player_age
\$score	Invalid in C/Java (<i>valid in JavaScript</i>)	score
total-marks	Invalid – - is treated as minus operator	totalMarks
userName	Valid	userName